1) $w = 16 \frac{\text{bits}}{\text{sample}}$

$r = 44100 \frac{\text{samples}}{s}$

$\Rightarrow \quad w \cdot r \cdot 2 = 16 \cdot 44100 \cdot 2 \frac{\text{bits}}{s} = 1411200 \frac{\text{bits}}{s}$

$\Rightarrow \quad 70 \text{ min} \cdot 60 \frac{s}{\text{min}} \cdot 1411200 \frac{\text{bits}}{s}$

$= 5927040000 \text{ bits}$

$= 706{,}56 \text{ MByte}$

2) $RMS = \frac{\text{magnitude}}{\sqrt{2}}$

$$\Rightarrow L = 20 \log_{10} \frac{\frac{0{,}1\ Pa}{\sqrt{2}}}{20\mu Pa} = 71{,}0\ dB\ SPL$$

$$L_{dB\,FS} = 10 \log_{10} \frac{P}{\frac{A^2}{2}}$$

$$P = \left(\frac{0{,}25}{\sqrt{2}}\right)^2$$

$$\Rightarrow L_{dB\,FS} = 10 \log_{10} \frac{\left(\frac{0{,}25}{\sqrt{2}}\right)^2}{\frac{1^2}{2}} = -12{,}0\ dB\,FS$$

$$a = L_{dB\,FS} - L_{dB\,SPL}$$

$$= -12{,}0 - 71 = -83\ dB$$

3) $-12\ dB\ 75 = 10 \log_{10} \frac{P_{Signal}}{\frac{A^2}{2}}$

$$P_{Signal} = \frac{A^2}{2} 10^{\frac{-12}{10}}$$

$$= 0{,}0315$$

$$P_{Noise} = \frac{\Delta^2}{12} \qquad \Delta = \frac{2A}{2^{en}}$$

$$= 77{,}61 \cdot 10^{-12}$$

$$SNR = 10 \log_{10} \frac{P_{Signal}}{P_{Noise}} = 10 \log_{10} \frac{0{,}0315}{77{,}61 \cdot 10^{-12}}$$

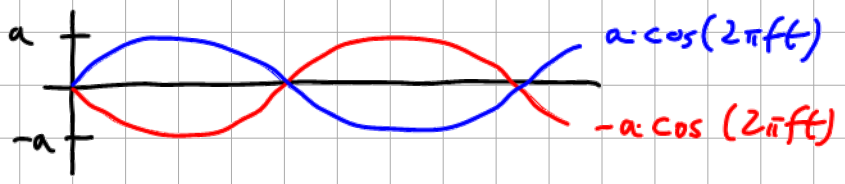
$$= 86{,}1\ dB$$

4) $RMS_x = \frac{a}{\sqrt{2}}$

$$75\,dB\,SPL = 20 \log_{10} \frac{RMS_x}{20\mu Pa} = 20 \log_{10} \frac{a}{\sqrt{2} \cdot 20\mu Pa}$$

$$\Rightarrow \quad a = \sqrt{2} \cdot 20\mu Pa \cdot 10^{\frac{75}{20}} = 0{,}159 \ Pa$$

The sign of a is undetermined:



Both signals have same RMS and same L.

The level is independent of the frequency: L is the same
L = 75 dB SPL

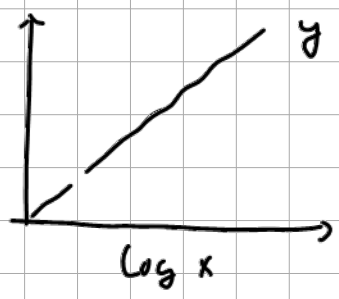
Exception: If the frequency is 0 Hz:

$$RMS = a$$

$$L = 20 \log_{10} \frac{a}{20\mu Pa}$$

$$\Rightarrow a = 20\mu Pa \cdot 10^{\frac{75}{20}} = 0{,}1125 \ Pa$$

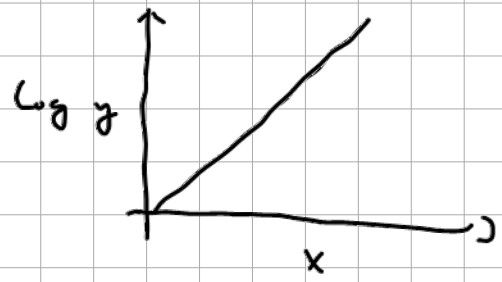
5) log x-axis:



$y = f(\log x) = m \cdot \log x$

$\Rightarrow$ f is a logarithmic function.

log y-axis:

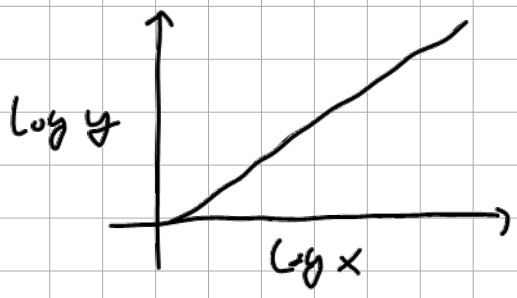


$\log y = \log f(x) = m \cdot x$

$y = e^{m \cdot x}$

$\Rightarrow$ f is an exponential function

log-log-plot



$\log y = \log f(\log x) = m \cdot \log x$

$y = e^{m \cdot x}$

$\Rightarrow$ f is a linear function.

6) Maximum digital magnitude $\hat{=}$ Overload point
$\hat{=}$ 0 dB FS

$$L_{dB\,FS} = a + L_{dB\,SPL}$$

$$0\ dB\ FS = -120\ dB + L_{dB\,SPL}$$

$$L_{dB\,SPL} = 120\ dB$$

7) $P = \frac{\Delta^2}{12} = \frac{1}{12} \frac{4A^2}{2^{2w}}$

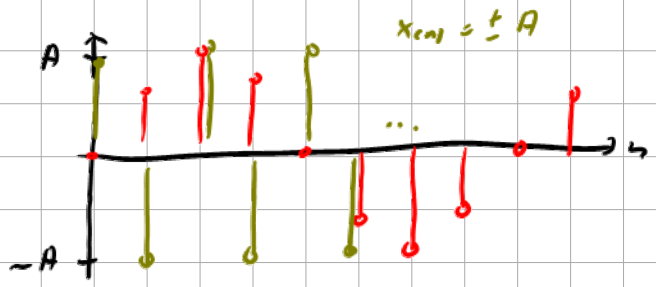
$\Delta = \frac{2A}{2^w}$

$L_{dB\ FS} = 10 \log_{10} \frac{2P}{A^2} = 10 \log_{10} \frac{2}{A^2} \cdot \frac{1}{12} \cdot \frac{4A^2}{2^{2w}} = 10 \log_{10} \frac{2}{3} \frac{1}{2^{2w}}$

$= 10 \log_{10} \frac{2}{3} \cdot \frac{1}{2^{32}} = -98{,}1$ dB FS

8) overload point : A

$x(n) = \pm A$



$y(n) \hat{=} 0$ dB FS

$$L = 10 \log_{10} \frac{P}{A^2/2} \qquad P_x = A^2$$

$$= 10 \log_{10} \frac{A^2}{A^2/2} = 3.01 \text{ dB FS}$$

dB FS is evaluated in digital domain.

The higher the magnitudes, the higher the dB FS.

Highest dB FS corresponds to highest possible magnitudes for each n.

The highest magnitude is A (= overload point).

$\Rightarrow \quad |x(n)| = A$ for all n.

$\boxed{\cancel{7}}$

9)

$$x(t) = \cos(7 \cdot t)^2$$

$$x(n) = x\left(\frac{n}{r}\right) = \cos\left(7 \cdot \frac{n}{10}\right)^2 \quad \text{for} \quad 0 \leq n \leq 5$$

$$x(0) = 1$$

$$x(1) = 0{,}585$$

$$x(2) = 0{,}029$$

$$x(3) = 0{,}255$$

$$x(4) = 0{,}988$$

```
import numpy as np

n = np.arange(5)
r = 10
x = np.cos(7*n/r)**2
```

10)

$$r_{min} = 2 \cdot f_c = 6800 \text{ Hz}$$

$$\frac{r}{r_{min}} = \frac{32000}{6800} = 4{,}7$$

$\Rightarrow$ downsampling by factor 4 possible

$f_c = 3400$ Hz corresponds to Narrowband speech

11)

- DC corresponds to the frequency $f = 0\,Hz$.

  This frequency is outside the passband of the bandpass.

  Therefore $z(n)$ has no DC component.

- Wideband speech has a frequency range of $50\,Hz \ldots 7000\,Hz$.

  $\Rightarrow$ $z(n)$ has nearly all components of wideband speech.

  $\Rightarrow$ Human voices are understandable in $z(n)$

- $y(n)$: $f_{max} = 24\,kHz$ (sampling theorem)

  $z(n)$: $f_{max} = 7\,kHz$

12) $x(t) = \cos(500\,t) = \cos(2\pi f_c \cdot t)$

$$f_c = \frac{500}{2\pi}$$

$$200 = r \geq 2 f_c = \frac{500}{\pi}$$

$$200 \geq \frac{500}{\pi} \checkmark$$ no alias occurs